

PRODUCT REQUIREMENT DOCUMENT (PRD)

*Multi-User Real-Time Convoy Navigation & Communication Platform
System Architecture & Implementation Matrix: Google Maps Platform vs. OpenStreetMap
Ecosystem*

1. Executive Summary & Core Value Proposition

This application specifies a comprehensive real-time, multi-user convoy navigation and communication ecosystem designed to synchronize travel across disparate vehicles. Traditional navigation platforms isolate individual drivers. This application bridges structural orchestration gaps by introducing unified group route frameworks, vehicle grouping matrix states, continuous peer-to-peer background location telemetry streaming, and zero-latency communication modules—consisting of duplex spatial conference calling and overriding push-to-talk (PTT) broadcast channels.

2. Comprehensive User Workflows

2.1 Trip Initialization & Deep Link Deployment

- **Authentication:** The session host orchestrator registers or signs in using secure OAuth2 or standard mobile verification wrappers.
- **Route Selection:** The host selects a target endpoint via an auto-suggest place lookup terminal. The system parses geography vector sets, requests transit paths from the active route engine, and overlays the optimal path polyline onto the local map container.
- **Session Generation:** Executing 'Create Convoy' triggers the platform backend to initialize a unique trip_id context coupled with security token tags inside the high-velocity data state layer.
- **Invitation Broadcast:** The application generates a deep-link path string containing specific runtime variables (e.g., ?trip_id=CONVOY_987X) which the host passes to target travellers via third-party wrappers (WhatsApp, SMS, Email).

2.2 Convoy Integration & Dynamic Onboarding

- **Universal Link Parsing:** An invited user launches the shared deep link. The system catches the intent, validates localized app installation flags, and redirects to the platform or app store seamlessly.
- **Dynamic Route Construction:** The application catches the raw trip_id parameters, queries the distributed backend cluster to fetch the destination coordinate footprint, and triggers turn-by-turn navigation updates relative to that passenger's distinct real-time GPS geolocation.

- **Roster Alignment:** Before tracking loop entry, the app forces a vehicle specification selection interface, forcing the target user to register their transit profile grouping (SUV, Sedan, Motorcycle, or Passenger class).
- **Active Sync Initialization:** The verified user automatically links into the background spatial polling sequence and launches on the master vehicle layout panels.

3. Detailed Functional Requirements

FR-1: Core Map Telemetry & Turn-by-Turn Engine

The map rendering runtime must plot coordinates, layer vector polyline segments, and evaluate turn guidance logic instantly. Lines must include explicit visual treatments detailing traffic conditions across the tracking path. Audio guidance modules must use localized Text-to-Speech (TTS) rules. Off-route deviations must trigger path recalculation within 2.5 seconds without breaking the active routing matrix profiles of neighboring convoy participants.

FR-2: Real-Time Multi-User Location Tracker

The platform must aggregate continuous active background location coordinates from all group participants. The mapping UI layout must render specialized custom map marker elements mapped to each participant user entity. Direction vector paths, immediate traveling velocity vectors, and active heading angles must stream coordinates across peer connections at uniform intervals ranging from 3 to 5 seconds maximum.

FR-3: Vehicle Roster Sidebar Matrix

The layout configuration must supply an expandable sliding navigation view indexing all active convoy entities classified by their vehicle choices. This layout updates connection heartbeats (In-Transit, Stopped, Offline), vehicle speed, and dynamic ETA outputs. Selecting a specific member within this roster must trigger map viewport canvas panning, instantly framing and focusing on that specific participant marker's current coordinates.

FR-4: Integrated Chat & Communication Module

The communication tier must accommodate isolated 1-on-1 text/voice chats, a group text room bound to the trip_id, a persistent WebRTC group audio channel, and a high-priority digital Walkie-Talkie Push-to-Talk (PTT) broadcast layer enabling any driver to blast overriding audio to all vehicles instantly.

4. Architectural Implementation Frameworks

Architects can validate implementation paths using the premium Google Maps ecosystem or the open-source OpenStreetMap (OSM) stack blueprint. The technical specifications below map how both environments achieve the target requirements.

Requirement Domain	Google Maps Platform Stack	OpenStreetMap (OSM) Ecosystem
Map UI Layer	Google Maps SDK (iOS & Android). Premium global vector graphics map delivery.	MapLibre Native SDK / Flutter Map. Renders open-source raster or vector data tiles free.
Route Engine	Google Routes API. Returns advanced routes containing accurate real-time traffic offsets.	OSRM (Open Source Routing Machine) or Valhalla. Custom-hosted logic running on cheap servers.
Navigation Guidance	Google Maps Navigation SDK (Premium Enterprise Access Key Required). Natively contained.	Mapbox Navigation SDK v1 (Open-source release branch). Embeds voice prompts natively.
Place Selection	Google Places API (Place Autocomplete lookup engine). Top-tier asset data pool.	Nominatim / Photon Index Search Engine. Open-source keyword-to-coordinate address translation.
Voice Processing	Agora Voice RTC Engine. Seamless cloud audio handling framework.	WebRTC SFU Cluster (mediasoup) with custom audio mixing configurations.

5. Non-Functional & Telemetry Performance Hardening

5.1 Mobile Device Battery Conservation Strategies

- **Stationary State (0 km/h):** Location polling drops to a 30-second heart-beat interval if velocity metrics register completely flat lines.
- **Urban Grid State (< 40 km/h):** Coordinate captures execute at 5-second loops or across 15-meter delta bounds to secure accurate intersection tracks.
- **Highway Speed State (> 40 km/h):** Telemetry updates accelerate to 3-second frequencies or 40-meter spatial distance offsets to sustain vector alignment.

5.2 Fluid Map Marker Geometric Interpolation

5.3 Voice Priority Matrix & WebRTC Override

6. Structural Convoy Database Schema Blueprint

```
{
  "convoys": {
    "trip_id_convoy_987x": {
      "session_metadata": {
        "host_user_id": "usr_alpha_11",
        "destination_coordinates": { "lat": 18.4842, "lng": 73.8943 },
        "destination_name": "Kondhwa, Pune, MH, India",
        "created_timestamp": 1784802660,
        "is_active": true
      },
      "routing_polyline_string": "a~l`Fizzy0_u@...[Encoded Path String]...",
      "active_vehicles": {
        "veh_suv_01": {
          "vehicle_label": "Fortuner (Lead)", "vehicle_type": "SUV",
          "assigned_occupants": [
            { "user_id": "usr_alpha_11", "name": "Aman (Driver)" },
            { "user_id": "usr_beta_22", "name": "Rahul" }
          ]
        }
      },
      "live_telemetry_stream": {
        "usr_alpha_11": {
          "current_position": { "lat": 18.5204, "lng": 73.8567 },
          "instantaneous_speed_kmh": 45.5, "heading_bearing_degrees": 185.2,
          "last_updated_time": 1784803120
        }
      },
      "communication_channels": {
        "webrtc_voice_bridge_token": "agora_rtc_token_string_abc123xyz",
        "active_broadcast_speaker_id": "usr_alpha_11",
        "active_broadcast_status": "TALKING"
      }
    }
  }
}
```